FLYBOAT

Guiyang High-Tech Yige Electronic Co., Ltd.

30S Series Metal Oxide Varistors — Datasheet

Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- For top grades the melt-through point is confined within a 0.5 cm radius of the chip centre, converting a random failure location into a controlled one.
- If the chip does melt through, the flame jet is shielded by the copper electrode, which conducts the heat to the trip point and drives the thermal disconnector open.
- Exceptional damp-heat resistance: under a customer's severe boil test (30 min in boiling water, cool, measure, re-boil; 6 cycles required with no significant change in varistor voltage or leakage), Yige product has reached 8 cycles with small-current parameters essentially unchanged.
- Surge and energy headroom (standard range): In up to 20 kA, Imax up to 40 kA, 2 ms energy absorption up to 1,580 J.
- Long, verifiable service life (Arrhenius law): verified minimum expected life of 110 years at 40 °C, 11 years at 65 °C, 0.7 years at 95 °C upper ambient temperature (115 °C / 1,000 h ageing basis).
- High TOV withstand: at a temporary overvoltage of 1.45 × Uc the chip stays thermally stable for more than 45 min (r.m.s.), with no damage.

Model naming: 30S K 431 — 30 = nominal size (mm), S = square; K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead/structure form; -U = uncoated bare electrode disc (no mark = epoxy-coated by default).

UL recognition: Five grades are UL 1449 Recognized under File E335803 per the UL Product iQ record: 621 / 681 / 821 / 112 / 122 (varistor voltage 620V–1200V) — models 30SK621, 30SK681, 30SK821, 30SK112 and 30SK122. The record also recognizes DC-rated variants of the 30S series, including three DC-only grades: 102 / 751 / 911 (varistor voltages 750–1,000 V). The catalog ratings tables below print no certification marks for the 30S series; the recognition stated here follows the UL Product iQ record for file E335803. How to verify: go to UL Product iQ (productiq.ulprospector.com) and search file number E335803. Full coverage for all series is on the Certifications page.

1. Electrical characteristics — standard range

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms).

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
30SK101	100	±10	60	85	20	40	360-3.6	150
30SK121	120	±10	75	100	20	40	400-3.6	185
30SK151	150	±10	95	125	20	40	450-3.0	230

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
30SK181	180	±10	115	150	20	40	490-2.8	270
30SK201	200	±10	130	170	20	40	520-2.6	310
30SK221	220	±10	140	180	20	40	550-2.6	350
30SK241	240	±10	150	200	20	40	600-2.6	370
30SK271	270	±10	175	225	20	40	680-2.6	430
30SK301	300	±10	195	250	20	40	770-2.6	480
30SK331	330	±10	210	270	20	40	830-2.55	520
30SK361	360	±10	230	300	20	40	910-2.55	580
30SK391	390	±10	250	320	20	40	1050-2.55	640
30SK431	430	±10	275	350	20	40	1150-2.55	730
30SK471	470	±10	300	385	20	40	1250-2.55	780
30SK511	510	±10	320	410	20	40	1300-2.55	840
30SK561	560	±10	350	450	20	40	1400-2.5	910
30SK621	620	±10	385	505	20	40	1550-2.5	990
30SK681	680	±10	420	560	20	40	1700-2.5	1000
30SK711	710	±10	440	585	20	40	1750-2.45	1010
30SK751	750	±10	460	615	20	40	1800-2.45	1020
30SK781	780	±10	485	640	20	40	1950-2.45	1030
30SK821	820	±10	510	670	20	40	2000-2.40	1080
30SK911	910	±10	550	745	20	40	2100-2.3	1170
30SK951	950	±10	575	765	20	40	2250-2.3	1180
30SK102	1000	±10	625	825	20	40	2300-2.3	1220
30SK112	1100	±10	680	895	20	40	2530-2.3	1350
30SK122	1200	±10	750	1060	20	40	2750-2.3	1400
30SK142	1400	±10	820	1120	20	40	3200-2.3	1450
30SK152	1500	±10	850	1240	20	40	3400-2.3	1500
30SK162	1600	±10	910	1320	20	40	3600-2.3	1540
30SK182	1800	±10	1000	1465	20	40	4000-2.3	1580

Notes: 30SK122’s UDC prints as 1060 V in the catalog — shown as printed.

2. Electrical characteristics — low-residual-voltage (low-clamping) range

Same column definitions as Table 1; optimised for lower clamping voltage at In.

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
30SK431	430	±10	275	350	20	40	950-2.2	730
30SK471	470	±10	300	385	20	40	1030-2.2	780
30SK511	510	±10	320	410	20	40	1130-2.2	840
30SK561	560	±10	350	450	20	40	1240-2.2	910
30SK621	620	±10	385	505	20	40	1350-2.15	990
30SK681	680	±10	420	560	20	40	1480-2.15	1000
30SK711	710	±10	440	585	20	40	1540-2.15	1010
30SK751	750	±10	460	615	20	40	1630-2.15	1020
30SK781	780	±10	485	640	20	40	1670-2.15	1030
30SK821	820	±10	510	670	20	40	1850-2.1	1080
30SK911	910	±10	550	745	20	40	1940-2.1	1170
30SK951	950	±10	575	765	20	40	2040-2.1	1180
30SK102	1000	±10	625	825	20	40	2200-2.1	1220
30SK112	1100	±10	680	895	20	40	2380-2.1	1350
30SK122	1200	±10	750	1060	20	40	2620-2.1	1400

3. Dimensions and thickness

Encapsulated thickness Tmax (mm), standard & low-clamping ranges — 101: 3.8 · 121: 3.8 · 151: 3.8 · 181: 3.9 · 201: 4.0 · 221: 4.1 · 241: 4.2 · 271: 4.4 · 301: 4.6 · 331: 4.7 · 361: 4.9 · 391: 5.1 · 431: 5.2 · 471: 5.5 · 511: 5.7 · 561: 6.0 · 621: 6.3 · 681: 6.3 · 711: 6.6 · 751: 7.1 · 781: 7.3 · 821: 7.5 · 911: 8.1 · 951: 8.1 · 102: 8.1 · 112: 8.9 · 122: 8.9 · 142: 9.9 · 152: 10.4 · 162: 10.9 · 182: 11.9.

Dimension drawings for each package form are available on request.